

**Tutorial 7 ONLY ALLOWED TEXT BOOK – NO WORK BOOKS
ALLOWED!**

Chapter 7 & 8: Kinetic Energy and Work, potential energy

9th Edition (*8th Edition*)

Monday:

1st session: Ch 7 Problem 19 , Ch 8 Problem 15 a, b

2nd session: Ch 7 Problem 5 , Ch 8 Problem 15 a, c

Tuesday:

Session 1: Chapter 7 Problem 57 a, b, Chapter 8 Problem 23 a

Session 2: Chapter 7 Problem 57 a, c, Chapter 8 Problem 23 b

Wednesday:

Session 1: Chapter 7 Problem 19, Chapter 8 Problem 15 a, b

Session 2: Chapter 7 Problem 57 a, b, Chapter 8 Problem 23 b

Solutions

Ch 7 # 5

(5)

A father racing his son has half the kinetic energy of the son, who has half the mass of the father. The father speeds up by 1.0 m/s and then has the same kinetic energy as the son. What are the original speeds of (a) the father and (b) the son?

We denote the mass of the father as m and his initial speed v_i . The initial kinetic energy of the father is

$$K_i = \frac{1}{2} K_{\text{son}}$$

and his final kinetic energy (when his speed is $v_f = v_i + 1.0$ m/s) is $K_f = K_{\text{son}}$. We use these relations along with Eq. 7-1 in our solution.

(a) We see from the above that $K_i = \frac{1}{2} K_f$ which (with SI units understood) leads to

$$\frac{1}{2} m v_i^2 = \frac{1}{2} \left[\frac{1}{2} m (v_i + 1.0 \text{ m/s})^2 \right].$$

The mass cancels and we find a second-degree equation for v_i :

$$\frac{1}{2} v_i^2 - v_i - \frac{1}{2} = 0.$$

The positive root (from the quadratic formula) yields $v_i = 2.4$ m/s.

(b) From the first relation above ($K_i = \frac{1}{2} K_{\text{son}}$), we have

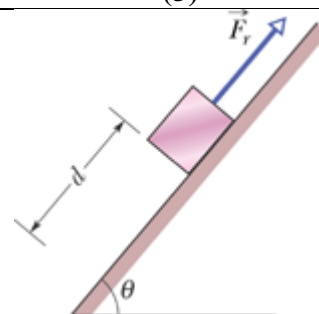
$$\frac{1}{2} m v_i^2 = \frac{1}{2} \left(\frac{1}{2} (m/2) v_{\text{son}}^2 \right)$$

and (after canceling m and one factor of 1/2) are led to $v_{\text{son}} = 2v_i = 4.8$ m/s.

#19

(5)

In Fig., a block of ice slides down a frictionless ramp at angle $\theta = 50^\circ$ while an ice worker pulls on the block (via a rope) with a force $\vec{F}_r$ that has a magnitude of 50 N and is directed up the ramp. As the block slides through distance $d = 0.50\text{ m}$ along the ramp, its kinetic energy increases by 80 J. How much greater would its kinetic energy have been if the rope had not been attached to the block?



$\Delta K = W_{net} = W_{app} + W_g$ applies

In the second situation there is no applied force, therefore the amount of Work done by the applied force would add to the amount of energy available as KE of the initial situation.

(This needs to be explained either in summary words or by using the equations for ΔK_i and ΔK_f) (the wording of the problem suggests that it is only necessary to examine the contribution from the rope (which would be the “ W_a ” term)):

$$W_a = F_r(d) \cos \theta = -(50 \text{ N})(0.50 \text{ m}) = -25 \text{ J}$$

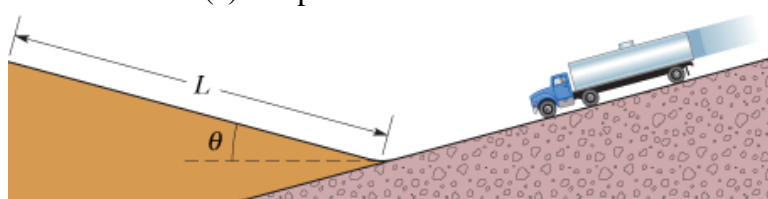
(negative because angle between $\vec{F}_r$ and $\vec{d}$ is 180°)

Thus, the kinetic energy would have been 25 J greater if the rope had not been attached (given the same displacement).

Ch 8 #15

(5)

In Fig. 8-36, a runaway truck with failed brakes is moving downgrade at 130 km/h just before the driver steers the truck up a frictionless emergency escape ramp with an inclination of $\theta = 15^\circ$. The truck's mass is $1.2 \times 10^4 \text{ kg}$. (a) What minimum length L must the ramp have if the truck is to stop (momentarily) along it? (Assume the truck is a particle, and justify that assumption.) Does the minimum length L increase, decrease, or remain the same if (b) the truck's mass is decreased and (c) its speed is decreased?



We neglect any work done by friction. We work with SI units, so the speed is converted: $v = 130(1000/3600) = 36.1 \text{ m/s}$.

(a) We use conservation of mechanical energy: $K_f + U_f = K_i + U_i$

With $U = 0$ at the bottom of the downhill we get:

$U_i = 0$, $U_f = mgh$ and $K_f = 0$. Since $K_i = \frac{1}{2}mv^2$, where v is the initial speed of the truck, we obtain

$$\frac{1}{2}mv^2 = mgh \Rightarrow h = \frac{v^2}{2g} = \frac{(36.1 \text{ m/s})^2}{2(9.8 \text{ m/s}^2)} = 66.5 \text{ m} .$$

If L is the length of the ramp, then $L \sin 15^\circ = 66.5 \text{ m}$ so that $L = (66.5 \text{ m})/\sin 15^\circ = 257 \text{ m}$. Therefore, the ramp must be about $2.6 \times 10^2 \text{ m}$ long if friction is negligible.

(b) The answers do not depend on the mass of the truck. They remain the same if the mass is reduced.

(c) If the speed is decreased, h and L both decrease (note that h is proportional to the square of the speed and that L is proportional to h).

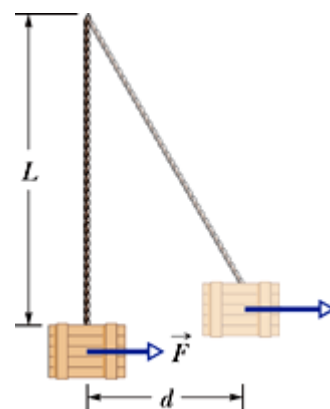
Tuesday:

Session 1: Chapter 7 Problem 57 a, b, Chapter 8 Problem 23 a

Session 2: Chapter 7 Problem 57 a, c, Chapter 8 Problem 23 b

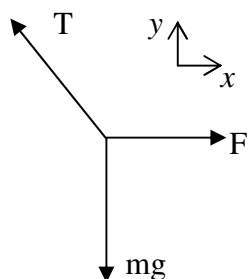
Chapter 7, # 57 a, b, c, d**(7)**

A 230 kg crate hangs from the end of a rope of length $L = 12.0$ m. You push horizontally on the crate with a varying force $\vec{F}$ to move it distance $d = 4.00$ m to the side (Figure). (a) What is the magnitude of $\vec{F}$ when the crate is in this final position? During the crate's displacement, what are (b) the total work done on it, (c) the work done by the gravitational force on the crate, and (d) the work done by the pull on the crate from the rope? (e) Knowing that the crate is motionless before and after its displacement, use the answers to (b), (c), and (d) to find the work your force $\vec{F}$ does on the crate. (f) Why is the work of your force not equal to the product of the horizontal displacement and the answer to (a)?



(a) To hold the crate at equilibrium in the final situation, $\vec{F}$ must have the same magnitude as the horizontal component of the rope's tension $T \sin \theta$, where θ is the angle between the rope (in the final position) and vertical:

$$\theta = \sin^{-1}\left(\frac{4.00}{12.0}\right) = 19.5^\circ.$$



$$\sum F_y = ma_y = 0 = T \cos \theta - mg$$

$$T = \frac{mg}{\cos \theta}$$

$$T = 2391 \text{ N}$$

$$\sum F_x = ma_x = 0 = F - T \sin \theta$$

$$F = T \sin \theta$$

$$F = 797 \text{ N.}$$

An alternative approach based on drawing a vector triangle (of forces) in the final situation

provides a quick solution.

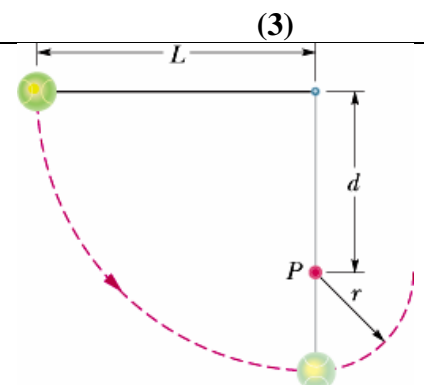
(b) Since there is no change in kinetic energy, the net work on it is zero.

(c) The work done by gravity is $W_g = \vec{F}_g \cdot \vec{d} = -mgh$, where $h = L(1 - \cos \theta)$ is the vertical component of the displacement. With $L = 12.0$ m, we obtain $W_g = -1547$ J which should be rounded to three figures: -1.55 kJ.

(d) The tension vector is everywhere perpendicular to the direction of motion, so its work is zero (since $\cos 90^\circ = 0$).

Chapter 8, # 23 a, b

The string in the Fig. is $L = 120$ cm long, has a ball attached to one end, and is fixed at its other end. The distance d from the fixed end to a fixed peg at point P is 75.0 cm. When the initially stationary ball is released with the string horizontal as shown, it will swing along the dashed arc. What is its speed when it reaches (a) its lowest point and (b) its highest point after the string catches on the peg?



(a) Let the lowest point have reference position of $y = 0$.

As the string reaches its lowest point, its original potential energy $U = mgL$ (measured relative to the lowest point) is converted into kinetic energy. Thus, from conservation of mechanical energy:

$$K_i + U_i = K_f + U_f$$

$$0 + U_i = K_f + 0$$

$$mgL = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2gL}.$$

With $L = 1.20$ m we obtain $v = 4.85$ m/s.

(b)

$$K_i + U_i = K_f + U_f$$

$$0 + U_i = K_f + U_f$$

Where the total mechanical energy in the final position is shared between kinetic $\frac{1}{2}mv_b^2$ and potential mgy_b . We note that $y_b = 2r$ where $r = L - d = 0.450$ m. Energy conservation leads to

$$mgL = \frac{1}{2}mv_b^2 + mgy_b$$

which yields $v_b = \sqrt{2gL - 2g(2r)} = 2.42$ m/s.

Wednesday:

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